REPORT
OF
AI-ML VIRTUAL INTERNSHIP
A Report on Artificial Intelligence and Machine Learning

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

Roll Number: CS-23411020
Name of Student: Keshav Narayan
Batch: 2023-27

2026-27

CANDIDATE'S DECLARATION

I, Keshav Narayan, Roll No. CS-23411020, student of Bachelor of Technology in Computer Science and Engineering, 7th Semester, Section 4CSE5, IILM University, Greater Noida, hereby declare that the internship report entitled "AI-ML Virtual Internship" is a record of my learning and work undertaken during the eight-week AI-ML Virtual Internship conducted through the AICTE–EduSkills National Internship Portal during April–June 2026. I hereby declare that the information presented in this report is prepared for academic purposes based on my internship learning, activities, assessments and project-related work. I have made sincere efforts to present the concepts and learning outcomes accurately.

Signature

Student Name: Keshav Narayan

Roll No.: CS-23411020

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout my eight-week AI-ML Virtual Internship. First and foremost, I am thankful to the AICTE–EduSkills National Internship Portal and EduSkills for providing me with the opportunity to participate in the AI-ML Virtual Internship. This internship provided me with valuable exposure to Artificial Intelligence and Machine Learning concepts and helped me understand their practical applications. I would also like to express my gratitude to Google for Developers and the organizations supporting the internship program for providing a structured learning environment and valuable technical resources. I sincerely thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me with the opportunity to undertake this internship as part of my B.Tech CSE curriculum. I am grateful to my faculty members for their guidance, encouragement and continuous support during the internship. Their suggestions helped me improve my technical understanding and approach towards solving problems. Finally, I would like to thank my parents, friends and everyone who supported me throughout the internship and encouraged me to complete the program successfully.

Signature

Student Name: Keshav Narayan

Roll No.: CS-23411020

INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Keshav Narayan

IILM University, Greater Noida

has successfully completed the 8-weeks

AI-ML Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By : **India Edu Program**

Google for Developers

Karthik Padmanabhan Developer Ecosystem Lead MENA & India, Google	Dr. Buddha Chandrasekhar Chief Coordinating Officer (CCO) AICTE, Ministry of Education	Shubhajit Jagadev Chief Executive Officer (CEO) EduSkills

Certificate ID : 4a67a7e2c7dc1a4cfa77

Student ID: STU67ebc8c856d251743505608

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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4. Project Description

4.1 Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have become important technologies in modern computer science and are being applied in areas such as healthcare, finance, education, transportation, cybersecurity, recommendation systems and automation. The AI-ML Virtual Internship provided an opportunity to develop a practical understanding of the fundamental concepts of Artificial Intelligence and Machine Learning. The internship was completed over a period of eight weeks during April–June 2026 through the AICTE–EduSkills internship program. The internship focused on understanding the basic principles of Artificial Intelligence and Machine Learning, data preparation, machine learning algorithms, model training, evaluation and the application of intelligent techniques to solve real-world problems. Machine Learning is a subset of Artificial Intelligence in which computer systems learn patterns from data and use those patterns to make predictions or decisions. During the internship, emphasis was placed on understanding how data is processed, how features are selected, how models are trained and how their performance can be evaluated. The internship also helped in developing practical programming and analytical skills. Working with datasets and machine learning workflows provided an understanding of how theoretical concepts can be converted into practical solutions. Overall, the internship helped bridge the gap between academic knowledge and practical application in the field of Artificial Intelligence and Machine Learning.

4.2 Organization Profile

The AI-ML Virtual Internship was undertaken through the AICTE–EduSkills National Internship Portal. EduSkills provides structured virtual internship programs designed to help students gain exposure to emerging technologies and industry-oriented skills. The AICTE–EduSkills internship ecosystem provides students with opportunities to learn through structured modules, assessments, practical activities and project-based learning. The AI-ML Virtual Internship is focused on developing foundational and practical knowledge in Artificial Intelligence and Machine Learning. The internship program is supported by industry and technology partners and is intended to improve the employability and technical capabilities of students. The internship certificate issued to me confirms the successful completion of the eight-week AI-ML Virtual Internship during April–June 2026.

4.3 Problem Statement

Traditional academic learning provides students with theoretical knowledge of Artificial Intelligence and Machine Learning, but practical exposure is equally important for understanding how these technologies are applied to real-world problems. One of the major challenges for students is understanding the complete machine learning workflow, starting from data collection and preprocessing and continuing through model development, training, evaluation and interpretation. Therefore, the primary purpose of this internship was to gain practical exposure to Artificial Intelligence and Machine Learning concepts and understand how machine learning techniques can be used to analyze data and develop intelligent solutions. The internship aimed to strengthen technical knowledge, programming skills, analytical thinking and problem-solving abilities through structured learning and practical activities.

4.4 Project Objectives

- The major objectives of the AI-ML Virtual Internship were:
- To understand the fundamental concepts of Artificial Intelligence and Machine Learning.
- To understand different types of Machine Learning, including supervised and unsupervised learning.
- To learn the basic steps involved in a Machine Learning workflow.
- To understand data preprocessing and preparation techniques.
- To study important Machine Learning algorithms used for prediction and classification.
- To understand model training and testing.

- To learn how Machine Learning models are evaluated using appropriate performance metrics.

4.5 Scope of the Project

The scope of the AI-ML Virtual Internship covered the fundamental and practical aspects of Artificial Intelligence and Machine Learning. The internship provided exposure to the complete basic Machine Learning workflow, including understanding the problem, preparing data, selecting relevant features, training models, testing models and evaluating their performance. The internship also helped in understanding the role of Python and commonly used data science and machine learning libraries in developing AI/ML solutions. The knowledge acquired during the internship can be applied to various domains such as: • Healthcare • Finance • Education • E-commerce • Cybersecurity • Recommendation Systems • Predictive Analytics • Image and Text Processing • Automation • Business Intelligence The internship was primarily focused on learning and practical implementation rather than large-scale production deployment.

4.6 Technologies and Tools Used

• **Technology / Tool**	• **Purpose**
• **Python**	• Programming and ML implementation
• **NumPy**	• Numerical computations
• **Pandas**	• Data manipulation and analysis
• **Matplotlib**	• Data visualization
• **Scikit-learn**	• Machine Learning algorithms and evaluation
• **Jupyter Notebook / Google Colab**	• Development and experimentation
• **Git**	• Version control
• **GitHub**	• Code and project management
• **Machine Learning**	• Prediction and classification
• **Artificial Intelligence**	• Intelligent problem solving

4.7 System Architecture

DATA COLLECTION

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DATA PREPROCESSING

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DATA ANALYSIS

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FEATURE SELECTION

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MODEL SELECTION

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MODEL TRAINING

↓

MODEL TESTING

↓

MODEL EVALUATION

↓

PREDICTION / OUTPUT

The general AI-ML workflow followed during the internship can be represented as a sequence of stages.First, relevant data is collected from an appropriate source. The collected data may contain missing values, duplicate records or inconsistent formats. Therefore, data preprocessing is performed to improve the quality of the dataset.After preprocessing, exploratory analysis is performed to understand the structure and patterns within the data. Relevant features are then selected for model development.Based on the nature of the problem, an appropriate Machine Learning algorithm is selected. The prepared dataset is divided into training and testing sets. The training data is used to train the model, while the testing data is used to evaluate its performance.Finally, suitable evaluation metrics are used to determine how well the model performs. The trained model can then be used to make predictions on new data.

4.8 Methodology

The internship followed a structured, module-wise methodology spread across eight weeks:

- Week 1 – Introduction to AI and ML: During the first week, the focus was on understanding the fundamentals of Artificial Intelligence and Machine Learning. Concepts such as AI, ML, types of Machine Learning and real-world applications were studied.
- Week 2 – Python for AI/ML: The second week focused on Python programming and its application in data analysis. Concepts related to NumPy, Pandas and basic data visualization were studied and practiced.
- Week 3 – Data Preprocessing: The third week focused on understanding datasets and preparing data for Machine Learning. Data cleaning, handling missing values, feature selection and data transformation were studied.
- Week 4 – Supervised Learning: The fourth week focused on supervised Machine Learning techniques. Regression and classification concepts were studied along with the basic working of commonly used algorithms.
- Week 5 – Unsupervised Learning: The fifth week introduced unsupervised learning techniques. The purpose of clustering and discovering patterns in unlabeled datasets was studied.
- Week 6 – Model Training and Evaluation: The sixth week focused on model training, testing and performance evaluation. Different evaluation measures were studied to understand how effectively a trained model performs.
- Week 7 – Practical AI/ML Applications: During the seventh week, the concepts learned during the internship were applied through practical activities and problem-solving exercises. The focus was on understanding how AI and ML techniques can be applied to real-world scenarios.
- Week 8 – Project, Assessment and Final Learning: The final week focused on completing the required practical work and assessments, reviewing the concepts learned throughout the internship and understanding the overall Machine Learning workflow.

4.9 Expected Outcomes

- The Understanding of Artificial Intelligence and Machine Learning fundamentals.
- Knowledge of supervised and unsupervised learning.
- Improved Python programming skills.
- Understanding of data preprocessing and data analysis. • Familiarity with Machine Learning algorithms.
- Understanding of model training and testing.
- Knowledge of Machine Learning model evaluation.
- Improved analytical and problem-solving abilities.
- Better understanding of real-world AI/ML applications.

The internship also helped me understand the importance of continuous learning in the rapidly developing field of Artificial Intelligence and Machine Learning.

4.10 Certificates of Completion

The Internship Completion Certificate issued for the **AI-ML Virtual Internship**, confirming the successful completion of the **eight-week internship during April–June 2026**, is attached in **Section 3 (Internship Completion Certificate)** of this report.

5. Bibliography / References

1. AICTE–EduSkills National Internship Portal,
   AI-ML Virtual Internship Program.

2. EduSkills Foundation,
   AICTE–EduSkills Virtual Internship Program.

3. Scikit-learn Documentation,
   Machine Learning algorithms and model evaluation.

4. Python Documentation,
   Python programming language and standard libraries.

5. NumPy Documentation,
   Numerical computing with Python.

6. Pandas Documentation,
   Data analysis and manipulation using Python.

7. Matplotlib Documentation,
   Data visualization using Python.

8. Google for Developers,
   AI and Machine Learning learning resources.

9. Course materials and learning resources provided during the AI-ML Virtual Internship.